

AI

1. Mathematics
 - 1.1. Linear Algebra
 - 1.1.1. Scalars
 - 1.1.2. Vectors
 - 1.1.3. Matrices
 - 1.1.4. Matrix Multiplication
 - 1.1.5. Dot Product
 - 1.1.6. Transpose
 - 1.1.7. EigenValues
 - 1.1.8. EigenVectors
 - 1.1.9. SVD (Singular Value De-Composition)
 - 1.1.10. Examples
 - 1.1.10.1. Prepresent Dataset as Matrix
 - 1.1.10.2. Word Embeddings as Vectors
 - 1.2. Probability & Statistics
 - 1.2.1. Mean / Median
 - 1.2.2. Variance
 - 1.2.3. Standard Deviation
 - 1.2.4. Probability Distribution
 - 1.2.5. Gaussian Distribution
 - 1.2.6. Conditional Probability
 - 1.2.7. Baye s Theorem
 - 1.2.8. Example
 - 1.2.8.1. Spam Detection Probability
 - 1.3. Calculus
 - 1.3.1. Derivates
 - 1.3.2. Partial Derivates
 - 1.3.3. Gradient
 - 1.3.4. Chain Rule
 - 1.3.5. Example
 - 1.3.5.1. Gradient Descent Optimization
2. Python
 - 2.1. Collections
 - 2.1.1. List, Tuple, Dictionary
 - 2.1.2. Functions
 - 2.1.3. Lambda
 - 2.1.4. OOP
 - 2.1.5. Modules and Packages
 - 2.1.6. Virtual Environments
 - 2.1.7. Decorators
 - 2.1.8. Generators
 - 2.1.9. Async Python
 - 2.2. Data Handling Libraries
 - 2.2.1. Numpy
 - 2.2.2. Pandas

- 2.2.3. Polars
 - 2.3. Visualization
 - 2.3.1. Matplotlib
 - 2.3.2. SeaBorn
 - 2.4. Tools
 - 2.4.1. Jupyter Notebook
 - 2.4.2. Google Colab
 - 2.4.3. VS Code Python
 - 2.5. Example Projects
 - 2.5.1. Analyze Netflix DataSet
 - 2.5.2. Data Cleaning using Pandas
 - 2.5.3. Visualization of Trends
 - 2.5.4. Build Mini Data analysis notebook
- 3. AI
- 4. Machine Learning
 - 4.1. Supervised Learning
 - 4.2. UnSupervised Learning
 - 4.3. Training Vs Testing
 - 4.4. Overfitting vs UnderFitting
 - 4.5. Bias Vs Variance
 - 4.6. Regression Algorithms
 - 4.6.1. Linear Regression
 - 4.6.2. PolyNominal Regeression
 - 4.6.3. Ridge Regression
 - 4.6.4. Lasso Regression
 - 4.6.5. Example Predict House Prices
 - 4.7. Classification of Algorithms
 - 4.7.1. Logistic Regression
 - 4.7.2. Decission Trees
 - 4.7.3. KNN
 - 4.7.4. Naïve Bayes
 - 4.7.5. Example Spam Detection
 - 4.7.6. Libraries
 - 4.7.6.1. Scikit-Learn
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 - 4.7.7.1. Random Forest
 - 4.7.8. Gradient Boosting
 - 4.7.9. XG Boost
 - 4.7.10. Clustering
 - 4.7.10.1. K-Means
 - 4.7.10.2. Hierarchical Clustering
 - 4.7.11. Example Customer Segmentation
- 5. Deep Learning
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 - 5.1.1. Perception
 - 5.1.2. Activation Functions

- 5.1.3. Forward propagation
 - 5.1.4. Back propagation
 - 5.2. Deep learning Frameworks
 - 5.2.1. PyTorch
 - 5.2.2. TensorFlow
 - 5.3. CNN (Computer Vision)
 - 5.3.1. Convolution Layers
 - 5.3.2. Pooling
 - 5.3.3. Example Image Classifier
 - 5.4. RNN(Sequence Model)
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 - 5.4.2. GRU
 - 5.4.3. Example Prediction
 - 5.5. Project
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 - 5.5.2. Image Classifier
- 6. NLP + LLM Foundation
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 - 6.1.1. Tokenization
 - 6.1.2. Stop Words
 - 6.1.3. Stemming
 - 6.1.4. TF-IDF
 - 6.2. Word Embeddings
 - 6.2.1. Word2Vec concept
 - 6.3. Transformer
 - 6.3.1. Attention Mechanism
 - 6.4. Example Sentiment Analysis
 - 6.5. Libraries
 - 6.5.1. Hugging Face Transformers
 - 6.6. Project Movie Review Sentiment Classifier
- 7. Gen AI
 - 7.1. Large Language Models
 - 7.1.1. GPT Architecture
 - 7.1.2. Token Embeddings
 - 7.1.3. Context Window
 - 7.2. Prompt Engineering
 - 7.2.1. Zero-shot prompting
 - 7.2.2. Few shot prompting
 - 7.2.3. Chain of thought prompting
 - 7.3. RAG
 - 7.3.1. Embedding
 - 7.3.2. Vector DataBase
 - 7.3.2.1. PineCode
 - 7.3.2.2. FAISS
 - 7.3.2.3. ChromaDB
 - 7.3.3. Projects

- 7.3.3.1. AI ChatBot
 - 7.3.3.2. Document Q&A system
 - 7.3.3.3. AI assistant
- 8. Model Deployment
 - 8.1. Fast API
 - 8.2. Docker
 - 8.3. AI API
 - 8.3.1. Open AI
 - 8.3.2. Anthropic
 - 8.3.3. Hugging Face API
 - 8.4. MLOps Basics
 - 8.4.1. Model Versioning
 - 8.4.2. Experiment tracking
 - 8.5. Tools
 - 8.5.1. MF Flow
 - 8.5.2. Weight & Biases
 - 8.6. AI System Design
 - 8.6.1. AI Pipe Line
 - 8.6.2. DATA + MODEL +API+APP
- 9. AI Coding Tools / IDE
- 10. Projects
 - 10.1. Resume Analyzer
 - 10.2. AI Medical chatbot
 - 10.3. AI code assistant
 - 10.4. AI document search engine.

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